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TABLE BASE ASSEMBLY AND KIT FOR ASSEMBLY INTO TABLE BASE

Abstract

A tabletop assembly having two base members that form an X with a post attached to the point in the X where the two base members cross. This assembly does not rotate because post is attached to the two base members at two places. The kit of this assembly can be more readily packaged and shipped cheaper than previously known assemblies.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority from U.S. Provisional Patent Application No. 63/561,7898 filed on 6 Mar. 2024, the teachings of which are incorporated herein.

BACKGROUND

[0002] This disclosure relates generally to the assembly of, and kits for the assembly of, furniture and, more particularly, to table bases and the assembly of, and kits for the assembly of, both bases and tables.

[0003] A so-called X-shaped table base is well-known and universally used in lunchrooms, restaurants, cafés, and the like. An X-shaped table base may have a single stamped or cast X-shaped bottom, a post that screws into the bottom with a threaded rod, and a top plate. Components may be formed of iron and/or steel, or of other materials. The table base may be attached to a tabletop, such as by using screws. Typically, assembly of the table includes placing the tabletop upside-down on a floor, inverting the X-shaped base and placing it onto the underside of the tabletop, and then securing the base to the top with the screws. As used herein, an “X-shaped” base may actually have a “+” shape, where the four horizontally-extending members of the bottom are spaced apart in ninety degree intervals.

[0004] The conventional X-shaped table base bottom is typically a single cast or stamp, and is stored and shipped as such, requiring a large box. For example, a thirty-inch X-base bottom may require a 32×32×2 box to accommodate protective packaging. Such packaging is inefficient, at least because it includes enclosure of a large amount of air space within the carton or package. As a result, a large cumulative amount of air is stored in a warehouse, packages are difficult to ship because they are large and not easily grasped and because the weight within a carton is concentrated in some members, impacting the structural integrity of the carton.

[0005] In most instance the base is shipped in a flat box and the post shipped in another box.

[0006] There is a need for an X-shaped table base assembly kit that stores and ships easily so as to place the base and the post in the one box, thus lowering packaging and logistic costs.

SUMMARY

[0007] Embodiments may provide modular assembly of an X-shaped table base, including components and assembly methods. Embodiments may provide a stable and strong X-shaped table base that stores and ships easily. Embodiments may provide ease of assembly of component parts into a sturdy structure that may provide registration of its four feet with respect to the orientation of a shaped tabletop. For example, a space having a number of tables may easily align the feet and tabletop with those of the other tables.

[0008] This specification discloses a table base assembly. The table base can comprise a bottom base having a bottom base bottom surface, a bottom base top surface, a bottom base longitudinal axis, a bottom base center member having a plurality of bottom base center member through-holes, a first bottom base linear member and a second bottom base linear member extending in opposite directions from the bottom base center member along the bottom base longitudinal axis, and a bottom base shoulder extending downwardly from the bottom surface.

[0009] It is further disclosed that the assembly has a top base having a top base bottom surface, a top base top surface, a top base longitudinal axis, a top base center member having a plurality of top base center member through-holes, a first top base linear member, a second top base linear member extending in opposite directions from the top base center member along the top base longitudinal axis, and a raised perimeter extending downwardly from the top base bottom surface except for opposed sides of the center member that could be substantially orthogonal to the top base longitudinal axis.

[0010] It is further disclosed that the assembly has a post with a post center longitudinal axis having a post top end and a post bottom end opposite the post top end.

[0011] It is also disclosed that a portion of the plurality of bottom base center member through-holes are aligned with a portion of the plurality of the top base center member through-holes when the top base bottom surface is placed onto the bottom base top surface and that the bottom end of the post is configured to be mounted on the top base top surface at the top base center member so that the post center longitudinal axis intersects with both the bottom base longitudinal axis and the top base longitudinal axis thereof at the same point.

[0012] It is further disclosed that the table base assembly may comprise a first plurality of thread members extending from the post bottom configured for being placed through the top base center member through-holes and the bottom base center member through-holes.

[0013] These at least two thread members may be threaded studs secured to the post bottom end and/or bolts.

[0014] It is further disclosed that that the post bottom end may include at least two post bottom end threaded holes and the at least two thread members are configured to be mated to a corresponding post bottom end threaded hole.

[0015] The specification further discloses that the table base assembly may comprise a plurality of nuts each configured for engaging threads of the at least two thread members.

[0016] It is further disclosed that the table base assembly may comprise at least two threaded rods configured for being secured to and extend outwardly from the post top end.

[0017] It is further disclosed that the table base assembly may further comprise a second plurality of threaded members each configured for being secured to and extend outwardly from the post top end.

[0018] The specification further discloses a tabletop mount platform having a perimeter, a bottom with a tabletop mount platform center member having at least two tabletop mount platform center member threaded through-holes, a tabletop mount platform planar top surface and a plurality of tabletop mount holes adjacent the perimeter, wherein the tabletop mount platform center member is configured to mate to the top end of the post.

[0019] It is further disclosed that at least two top threaded rods may be configured to matingly engage with the at least two tabletop mount platform center member threaded through-holes.

[0020] There may also be a tabletop mount platform having a perimeter, a bottom, a center member having a plurality of threaded through-holes, a top configured for attaching a planar surface thereto, and a plurality of mounting holes adjacent the perimeter, wherein the tabletop mount platform center member is configured to be mated to the post top end.

[0021] The table base assembly may further comprise a second plurality of threaded members attached to a post top end configured to matingly engage with the threaded holes of the tabletop mount platform.

[0022] The table base assembly may have a plate attached to an inside surface of the post, the plate having an outwardly to facing surface that includes a plurality of holes and that can be substantially orthogonal to a lengthwise axis of the post.

[0023] It is also disclosed that the top base center member may have a plurality of top base mate members and the bottom base center member has a plurality of bottom base mate members and the plurality of top base mate members are configured to interact with the bottom base mate members to prevent rotation of the top and bottom base center member around the post bottom end and align the top base center member through-holes with the bottom base center member through-holes.

[0024] A kit for assembly into the tabletop base is also disclosed.

[0025] It is to be understood that the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF DRAWING FIGURES

[0026] Advantages and principles of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures in which:

[0027] FIG. **1** is a top view of components of a table base assembly, according to an exemplary embodiment;

[0028] FIG. **2** is a bottom perspective view of the table base assembly of FIG. **1**;

[0029] FIG. **3** is a perspective view of an exemplary assembly of two base members;

[0030] FIG. **4A** is a view of a first threaded rod.

[0031] FIG. **4B** is a perspective view of a nut.

[0032] FIG. **4C** is a view of a second threaded rod.

[0033] FIG. **5** is a perspective view of an assembled table base, ready to be attached to a tabletop, according to an exemplary embodiment;

[0034] FIG. **6** is a perspective view of a cover, in an exemplary embodiment;

[0035] FIG. **7** is a perspective top view of an assembled table base, according to an exemplary embodiment;

[0036] FIG. **8** is perspective view of a member of a table base assembly, according to an exemplary embodiment;

[0037] FIG. **9** is a schematic end view of a first plate that may be installed inside a post, according to an exemplary embodiment; and

[0038] FIG. **10** is a schematic end view of a second plate that may be installed inside a post, according to an exemplary embodiment.

[0039] FIG. **11A** shows a blow up view of a base foot mount on the base top.

[0040] FIG. **11B** shows a straight on view of a base foot mount.

[0041] FIG. **12** shows an exemplary kit in its package.

[0042] FIG. **13** shows an exemplary kit with the post outside its package.

[0043] FIGS. **14A**, **14B**, and **14C** show different views of a base foot.

[0044] FIG. **15** shows screws used to attach a tabletop to the assembly.

[0045] Corresponding reference characters indicate corresponding or similar parts throughout the several views.

DETAILED DESCRIPTION

[0046] Reference will now be made to the various Figures in which, unless otherwise noted, like numbers refer to like structures. As described herein and in the claims, the following numbers refer to the following structures as noted in the Figures. [0047] **10** refers to a post. [0048] **11** refers to a post center longitudinal axis. [0049] **20** refers to a bottom base. [0050] **30** refers to a top base. [0051] **40** refers to threaded studs extending from the post bottom end also considered bottom thread members. [0052] **50** refers to a post bottom end. [0053] **60** refers to threaded studs extending from the post top end, also considered post top end thread members. [0054] **70** refers to a post top end. [0055] **80** refers to a first bottom base linear member. [0056] **90** refers to a second bottom base linear member. [0057] **100** refers to a bottom base longitudinal axis. [0058] **110** refers to a bottom base center member. [0059] **120** refers to a bottom base center member through-holes. [0060] **130** refers to bottom base mate members, or nubs. [0061] **140** refers to a bottom base perimeter member. [0062] **150** refers to bottom base shoulders. [0063] **160** refers to a bottom base center member side. [0064] **170** refers to a bottom base center member side. [0065] **180** refers to a first top base linear member. [0066] **190** refers to a second top base linear member. [0067] **200** refers to a top base longitudinal axis. [0068] **210** refers to a top base center member. [0069] **220** refers to a top base center member through-holes. [0070] **230** refers to top base mate members, or nubs. [0071] **240** refers to a top base perimeter member. [0072] **250** refers to a first top base gap.

[0073] **260** refers to a second top base gap. [0074] **270** refers to a bottom base bottom surface. [0075] **280** refers to base feet. [0076] **282** refers to base feet bolts. [0077] **285** refers base feet pads. [0078] **286** refers to base feet mounts. [0079] **287** refers to base feet mount holes. [0080] **289** refers to screws. [0081] **290** refers to a first top base shoulder. [0082] **300** refers to a second top base shoulder. [0083] **310** refers to a top base bottom surface. [0084] **315** refers to a top base top surface. [0085] **320** refers to a bottom base top surface. [0086] **330** refers to a first top base shoulder edge. [0087] **340** refers to a second top base shoulder edge. [0088] **410** refers to a tabletop mount platform. [0089] **420** refers to a tabletop mount platform bottom surface. [0090] **430** refers to a tabletop mount platform center member. [0091] **440** refers to a tabletop mount platform hole. [0092] **450** refers to a tabletop. [0093] **460** refers to a cover. [0094] **470** refers to a cover center hole. [0095] **480** refers to cover notches. [0096] **490** refers to a cover top surface. [0097] **500** refers to a first cover notch side. [0098] **510** refers to a second cover notch side. [0099] **520** refers to a cover side member. [0100] **530** refers to a tabletop mount platform center member through-hole. [0101] **540** refers to a tabletop mount platform planar top surface. [0102] **550** refers to a first plate. [0103] **560** refers to first plate attachment members. [0104] **570** refers to a first plate holes. [0105] **580** refers to a first plate outwardly-facing surface. [0106] **590** refers to a tabletop mount platform mount cavity. [0107] **600** refers to a second plate. [0108] **610** refers to a second plate attachment member. [0109] **620** refers to a second plate outwardly-facing surface. [0110] **630** refers to second plate holes. [0111] **640** refers to a second plate center hole. [0112] **1330** refers to a bolt. [0113] **1340** refers to a bolt head. [0114] **1350** refers to bolt threads. [0115] **1360** refers to a nut. [0116] **1370** refers to a nut threads. [0117] **1380** refers to a threaded rod. [0118] **1390** refers to a bolt head. [0119] **1400** refers to threads on the threaded rod. [0120] **2000** refers to a packaging box. [0121] **2100** refers to packaging fill. [0122] **2200** refers to a hardware bag.

[0123] The embodiments described below are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of these teachings.

[0124] In a strong table base, structural elements may be provided, for example, as cast or stamped iron or steel parts, and may generally be thick in order to provide a heavy duty and secure assembly that may be utilized without the possibility of wobble, deformation, or weakness. Any other materials may alternatively be used. Component parts may be painted by any method including, but not limited to, powder, liquid, paste and other methods, whereby finished components may be protected against oxidation and may have increased thickness, aesthetics, strength, durability, adhesion properties, and others. Surface treatments may include any of anodizing, hot dipping, electroplating, chemical processes, enameling, painting, electrophoretic deposition, chemical agent resistant coating, and others.

[0125] FIG. 1 is a top view of components of a table base assembly that includes a post (**10**) and a two-piece X-shaped base that includes a bottom base (**20**) and a top base (**30**), according to an exemplary embodiment.

[0126] The table base assembly may also include threaded members to hold the top base, bottom base, and post together. These thread members which may be selected from the group consisting of threaded studs or studs (**40**), bolts (**1330**) and threaded rods. which by definition have bolt threads (**1350**) and may have a bolt head (**1340**), or threaded rods (**1380**) which have no head on either end. At least two thread members may be used at the post top end (**70**) known as two top thread members and at least two threaded rods may be used at the post bottom end (**50**) known as two bottom thread members and at least two threaded rods may be used at the post top end (**70**). The use of at least two threaded rods at the bottom end of the post prevents the base from rotating about the post bottom. The use of two threaded rods at the post top prevents the table mount from rotating about the top end of the post.

[0127] Post (**10**) may be formed of a thick gauge steel as a hollow tube with a structure near each

of its distal ends configured for receiving or projecting one or more thread members such as a stud, a bolt or a rod. As shown in FIG. 1, post (10) may include a first plurality of threaded studs (40) extending from the post bottom end (50) known as bottom thread members, and/or a second plurality of threaded studs (60) extending from the post top end (70) known as post top end thread members. In one embodiment, there are at least two post top end thread members. These threaded studs may be attached to the top post end and/or the bottom post end. The post may also have post bottom end thread holes for the introduction of a bolt or threaded rod.

[0128] In one embodiment one end of the threaded studs is attached to a structure, such as a plate or crossbars, at the bottom end of the post and optionally the top end of the post. This attachment can be done by welding, milling, or threading one end of the stud into threaded hole. Welding is an example of a permanent attachment.

[0129] As shown in FIG. 5, the studs are placed through, or pass through, the holes of the top base center member (210) and bottom base center member (110) and are then secured with nuts which may use a lock washer or nylon threads. In this instance, a user will pass the studs through the top base center member through-holes (220) and the bottom base center member through-holes (120). Nuts are then placed on the threaded portions of the studs.

[0130] In another embodiment the thread members are attached to corresponding bottom post end threaded holes and optionally a corresponding post top end threaded hole. In this instance, a user will use bolts to pass through the bottom base center member through-holes (120) and the top base member through-holes (220) and screw them into the threaded holes in the post.

[0131] The post also has a center longitudinal axis (11) that passes through the center of the post along the length of the post as shown in FIG. 1.

[0132] Bottom base (20) includes a first bottom base linear member (80) and second bottom base linear member (90) which extend away from one another in opposite directions along a bottom base longitudinal axis (100). Bottom base (20) may have a bottom base center member (110) that includes two through-holes (120) and four bottom base mate/raised members, or nubs (130). In various embodiments, any number of bottom base center member through-holes (120) and bottom base mate members (130) may be provided. In the illustrated embodiment, bottom base mate/raised members (130) may have a dome shape, but in various embodiments, bottom base mate members (130) may have any shape. Bottom base (20) may include a bottom base perimeter member (140) with bottom base shoulders (150) having a similar or same profile along both sides, including the sides of the bottom base center member sides (160) and (170).

[0133] Top base (30) includes a first top base linear member (180), and a second top base linear member (190) that extend away from one another in opposite directions along a top base longitudinal axis (200). Top base (30) may have a top base center member (210) that includes at least two top base center member through-holes (220).

[0134] Top base (30) may further have a top base top surface (315) and a top base bottom surface (310) opposite the top base top surface.

[0135] Top base (30) may include a top base perimeter member (240) that has a same, or similar, profile along its sides, except for top base gaps (250), (260) along opposed sides of the top base center member (210).

[0136] FIG. 2 is a bottom perspective view of the table base assembly of FIG. 1. Bottom base (20) may have bottom base shoulders (150) that are raised perimeter members extending downwardly from the bottom base bottom surface (270) along both lengthwise sides of bottom base (20).

[0137] Base feet (280) may be disposed at distal ends of bottom base (20) to provide resting surfaces for the table base assembly when it is assembled and installed. Base feet may include, for example, rubber, felt, wood, steel, plastic, or other material, and may be provided as components that are secured to the distal ends using a well-known threaded rod and threaded receptacle structure (not shown), allowing a user to level the assembled table by rotating individual feet. Top base (30) includes base feet (280) with a same structure.

[0138] Exemplary base feet are shown in FIGS. 14A, 14B, and 14C. The exemplary base foot has a base foot pad (285) that is attached to base foot bolt (282) which thread into a base foot mount hole (287) of a base foot mount (286) (See FIGS. 11A and 11B). As all four base feet are threaded, the base feet length extending from the respective base can be varied to level the tabletop.

[0139] Top base (30) may have a first top base shoulder (290) and a second top base shoulder (300) that are raised top base perimeter members (240) extending downwardly from a top base bottom surface (310) along both lengthwise sides of top base (30), except for a first top base gap (250) and a second top base gap (260) on opposed sides of the top base center member (210).

[0140] As shown in the embodiment in FIG. 2, top base center member (210) of top base (30) has four top base mate members (230). In various embodiments, any number of top base mate members (230) may be provided. In the illustrated embodiment, top base mate members (230) may have an inverted dome shape (nubs) but, in various embodiments, top base mate members (230) may have any shape. In various embodiments, the shapes of the top base mate members (230) are configured to interact with the bottom base mate members (130) for alignment purposes.

[0141] In one embodiment, top base mate members are concave up from the top base bottom surface and the bottom base mate members are concave up from the bottom base top surface and located so that bottom base mate members (130) may be snugly inserted into top base mate members (230). This interaction serves to align the through-holes of the top base with the through-holes of the bottom base.

[0142] Another embodiment of the top base mate member may be where the top base mate member is a hole and the bottom base mate member is a post, a protrusion, or nub, protruding upward from the bottom base top surface. Alternatively, the top base mate member may be a post protruding downward from the top base bottom surface with the bottom base mate member being a hole to receive the top base mate member.

[0143] In assembling the table base, in an exemplary embodiment, the post, the top base, and the bottom base are aligned so that the post center longitudinal axis will intersect with the top base longitudinal axis and the bottom base longitudinal axis and at least two of the threaded members, in one case the threaded studs (40) of post (10) are inserted into and through the top base center member through-holes (220), and into and through the bottom base center member through-holes (120). Bottom base mate members (130) on the bottom base top surface (320) of bottom base (20) thereby interact with top base mate members (230) on the top base bottom surface (310).

[0144] When assembling the table base, the top base is placed upon the bottom base so that an angle formed by the top base longitudinal axis and the bottom base longitudinal axis is in a range of 1 Degree to 89 Degrees. More preferably, this angle is within the range of 15 Degrees and 75 Degrees, with a range of 30 Degrees to 60 Degrees even more preferred and with a range of 40 Degrees to 50 Degrees most preferred.

[0145] Additionally, at least a portion of the plurality of bottom base center member through-holes are to align with at least a portion of the plurality of the top base center member through-holes when the top base bottom surface is placed onto the bottom base top surface

[0146] Nuts, as shown in FIG. 4B, described herein below, may be fastened onto the ends of threaded studs (40) and tightened, whereby top base (30) and bottom base (20) are secured to post (10) and thereby form a table base.

[0147] In an exemplary embodiment, washers (not shown), such as lock washers and flat washers, and/or thread-tightening fluid may be used to securely fasten the table base assembly together.

[0148] In an exemplary embodiment of an assembled table base, top base longitudinal axis (200) is orthogonal to bottom base longitudinal axis (100). Engagement/interaction of the bottom base mate members (130) with top base mate members (230), together with the use of one or more fasteners such as nuts and bolts, assures proper registration of the table base assembly. In this manner, the top base and bottom base linear members do not rotate about the longitudinal axis of the post.

[0149] FIG. 3 is a perspective view of an exemplary assembly of top base (30) onto bottom base

(20), where bottom base mate members (130) (e.g., FIG. 1) become engaged and interact with the top base mate members (230) (e.g., FIG. 2), and bottom base center member and top base center member through-holes (120) and (220), respectively, thereby become aligned with one another. As shown by example, a first top base shoulder edge (330) is thereby made to be in close proximity to bottom base center member side (160) of top base center member (110), and second top base shoulder edge (340) is made to be in close proximity to bottom base center member side (170) of center member (110). In the same manner, for example, shoulder edges (not shown) of the other side of top base (30) are respectively made to be in close proximity with bottom base center member sides (160), (170).

[0150] FIG. 4A is a view of a type of thread member called a bolt (1330). Bolt (1330) will have a bolt head (1340) and bolt threads (1350).

[0151] FIG. 4B is a perspective view of a nut (1360) having nut threads (1370).

[0152] FIG. 4C is a view of a threaded rod (1380) which is another type of threaded member. Threaded rod (1380) will have threads (1400). Not all of the threaded rod, bolt or stud may have threads. These thread members include portions with threads and bolt portions without threads.

[0153] In some embodiments, threaded rods (1380) (e.g., FIG. 4C), extending from the post bottom end, may pass through respective through-holes (220), (120) of top base (30) and bottom base (20) and be secured to bottom base (20) using nuts (1360). Washers (not shown), for example lock washers and flat washers, and thread-tightening fluid may also be used for assuring a strong and secure attachment of post (10).

[0154] FIG. 5 is a perspective view of an assembled table base with a tabletop mount platform (410), ready to be attached to a tabletop, according to an exemplary embodiment.

[0155] A tabletop mount platform (410) may be provided with a tabletop mount platform bottom surface (420) having a tabletop mount platform center member (430) formed to provide a level surface for mounting post (10) thereto. A number of tabletop mount platform holes (440) may be provided through tabletop mount platform (410) for securing tabletop mount platform (410) to the bottom surface of a tabletop (450) as shown in FIG. 7. Base feet (280) may be provided at distal ends of top base (30) and bottom base (20), and may be adjustable in height, for leveling the assembled table after it is fully assembled and placed into a user location.

[0156] FIG. 7 is a perspective view of an assembled table base, according to an exemplary embodiment. Tabletop mount platform (410) will have a minimum of two tabletop mount platform center member through-holes but may include any number of tabletop mount platform center member through-holes (530) for receiving threaded portions extending out of post top end (750). Various nuts and washers, for example lock washers and flat washers, and thread-tightening fluid may be used for securing mounting platform (410) to post (10). Tabletop mount platform (410) may preferably have a tabletop mount platform planar top surface (540) for attachment to a bottom surface of a tabletop (450).

[0157] As shown in FIG. 7, the tabletop mount platform will have a tabletop mount cavity formed by the tabletop mount platform planar top surface and the tabletop center member mount platform. The threaded portion extending past the post top end must be shorter than the distance from the tabletop center member and the tabletop mount platform planar top surface or alternatively, do not extend past the tabletop mount platform surface.

[0158] In various embodiments, attachment of a tabletop to a table base may utilize any fastening device(s) (not shown) including, but not limited to, screws, bolts, spacers, washers, cam pins/locks, pins, dowel rods, receptacle bracket(s) and lever-type engagement members, and others.

[0159] The bottom surface of a tabletop (450) may include pre-drilled holes and/or inserts for receiving a fastening device, whereby proper registration and alignment of tabletop (450) respecting the position of the “+” of the table base is assured. For example, in a restaurant having a number of identical tables, it may be desirable to provide the tables in a configuration where the “+” of all table bases are aligned with one another, and where all tabletop shapes (e.g., square,

rectangle, etc.) are also aligned with one another. In an embodiment, such registration may be provided by use of a plurality of thread members such as studs, bolts or threaded rods and use of tabletops having appropriate preset mounting structure (e.g., pre-drilled holes) on the bottom surface of the tabletop.

[0160] FIG. 6 is a perspective view of a cover (460), in an exemplary embodiment. Cover (460) may have a cover center hole (470) and a number of cover notches (480). Cover center hole (470), for example, may have a diameter slightly larger than the diameter of post (10). During assembly of a table base, cover (460) may be slipped onto post (10) with cover top surface (490) facing the post top end (70) of post (10). After securement of bottom base (20) and top base (30) to post (10), cover (460) may be pushed down on top of such secured base members, where cover notches (480) are oriented and sized to fit over respective underlying members of bottom base (20) and top base (30). For example, cover notches (480) may be provided at ninety-degree intervals, where each notch (480) may include a first cover notch side (500) and a second cover notch side (510) that are shaped to complement a shape of a corresponding underlying member. Cover side members (520) of cover (460) may include protrusions (not shown) that allow cover (460) to be snapped into place, for example by use of such protrusions along bottom edges of cover side members (520) and/or along inner surfaces of cover side members (520). Such protrusions, for example, may be formed to engage, interact, or mate, with edges and/or hollow members along corresponding underlying members of bottom base (20) and top base (30). As a result, and as shown further hereinbelow, cover (460) may be provided to hide the intersection of bottom base (20) with top base (30), thereby providing an enhanced appearance and, in some embodiments, structural support.

[0161] FIG. 8 is a perspective view of a member of a table base assembly showing an embodiment using cover (460) to provide a more aesthetically pleasing appearance by placing cover (460) over the intersecting members of top base (30) and bottom base (20), especially along top base gaps (250), (260) (e.g., FIGS. 1-15). An assembly method for a table base may include placing cover (460) over post (10) before attachment of top base (30) and bottom base (20) to post (10). Cover (460) may slide along post (10) and be secured by a snap fit, or by use of any other structure and method.

[0162] An alternative mounting structure is shown in FIG. 9. FIG. 9 is a schematic end view of a first plate (550) that may be installed inside post (10), according to an exemplary embodiment. First plate (550) may have any number of first plate holes (570), provided there are at least two first plate holes. First plate (550) may be attached, at a distance from the post bottom end (50), to the inside of post (10) using first plate attachment members (560). First plate (550) may also/alternatively be attached, at a distance from top end (70) of post (10), to the inside of post (10) using attachment members (560). Attachment members (560) may include, but are not limited to, L-shaped brackets, convex mate members, concave mate members, welds, etc. For example, at least three projections may be formed at separate locations along an imaginary plane that is orthogonal to the longitudinal center axis of post (11), first plate (550) may be placed to abut such projections, and first plate (550) may then be spot welded to such projections, whereby first plate outwardly-facing surface (580) is orthogonal to the post longitudinal center axis (11). In an embodiment, first plate (550) may be secured in a desired location without welding. In an embodiment, first plate (550) may have first plate holes (570) that may be either through-holes or threaded holes. In an embodiment, threaded bolts (1330) (e.g., FIG. 4A) and/or threaded rods (1380) (e.g., FIG. 4C) may have a same thread pitch and size as that of first plate holes (570), and may be used for securing bottom base (20) to post (10). In another embodiment the thread members are threaded into the corresponding bottom post end threaded holes and optionally a corresponding post top end threaded hole.

[0163] In an embodiment, threaded bolts (1330) and/or threaded rods (1380) may be used for securing the tabletop mount platform (410) to post (10). In an example embodiment, bolt (1330)

may be fixedly attached to first plate hole (570), such as by welding or otherwise adhering bolt (1330) to hole (570). In such a case, bolt (1330) may be threaded/fastened into an inwardly-facing surface (not shown) of plate (550), and bolt head (1340) may be welded to first plate (550) before installing first plate (550) into post (10), whereby bolt (1330) extends out of post bottom end (50), and/or post top end (70) depending on the embodiment of post (10). In another example, bolt (1330) may be inserted into washers, when used, into aligned top and bottom base center member through-holes (120), (220), and then fastened into first plate hole (570) in an embodiment where first plate hole (570) is threaded and where bolt head (1340) of bolt (1330), and any washers, are adjacent bottom base bottom surface (270). In embodiments, variations for use of first plate (550) may be implemented for installation of first plate (550) near post top end (70), whereby tabletop mount platform (410) may be secured to the post (10) by being secured to a first plate (550) disposed therein.

[0164] In an embodiment, a first, first plate (550) may be installed near post bottom end (50), a second first plate (550) may be installed near post top end (70), first plate holes (570) of both such first plates (550) may be through-holes, and threaded rod (380) may be inserted into optional washers, through through-holes (120), (220), through a first plate through-hole (570), through a second first plate through-hole (570), and through a table mounting platform center member through-hole (530), whereby a threaded end of threaded rod (1380) extends from an tabletop mount platform mount cavity (590). In such a case, a nut (360) may be tightened onto the exposed threads (400) of threaded rod (1380) extending from inner top surface (590), whereby bolt head (1340) becomes tightened against bottom base bottom surface (270), and a single long threaded rod (1380) may be used to form an assembled table base. In this embodiment, a use of two or more threaded rods (1380), and a corresponding number of holes along the insertion path of threaded rod (1380), improves the strength and stability of an assembled table base. Any combination of the above-described embodiments is envisaged.

[0165] FIG. 10 is a schematic end view of a second plate (600) that may be installed inside post (10), according to an exemplary embodiment. Second plate (600) may be attached, at a distance from post top end (70) of post (10), to the inside of post (10) using second plate attachment members (610), as an alternative to first plate (550). Second plate (600) may also/alternatively be attached, at a distance from post bottom end (50) of post (10), to the inside of post (10) using attachment members (610). Attachment members (610) may include, but are not limited to, L-shaped brackets, convex mate members, concave mate members, welds, etc. For example, at least three projections may be formed at separate locations on the inside surface of post (10), along an imaginary plane that is orthogonal to the longitudinal axis of post (10), second plate (600) may be placed to abut such projections, and second plate (600) may then be spot welded to such projections, whereby second plate outwardly-facing surface (620) is orthogonal to the post center longitudinal axis (11). Second plate (600) may have any number of second plate holes (630) that may be either through-holes or threaded holes. Installation of second plate (600) may be performed in a manner similar to that for an installation of a first plate (550). In an embodiment, second plate (600) may have a round shape with a diameter slightly less than a diameter of post (10). In an embodiment that uses L-shaped brackets as attachment members (610), a member of such L-bracket may have a rounded shape that matches the round shape of the inside surface of post (10), whereby a surface of the L-bracket having such rounded shape may be easily welded to the post inside surface to thereby provide a second plate attachment member (610). Any appropriate number and position of second plate holes (630) may be provided on second plate (600). In an embodiment, two second plate holes (630) may be provided, for example, as through-holes in alignment with threaded rods (1380) extending from post bottom end (50), and/or post top end (70).

[0166] The second plate may also contain a second plate center hole. (640).

[0167] In another embodiment, a single, threaded, second plate center hole (640) may be provided in second plate (600) and a single threaded rod (1380) may be fixedly attached to second plate

(600) to thereby extend from post top end (70). In such an embodiment, for example, tabletop mount platform (410) may be configured to have a threaded, centered hole (not shown) that may receive the single threaded rod, whereby a tabletop attached to such a tabletop mount platform (410) may be secured to or removed from a table base, by simply spinning such tabletop until it is tightened (e.g., clockwise) or made free (e.g., counterclockwise), respectively. This spinning embodiment allows a user to remove a tabletop, including an attached tabletop mount platform, from a table base, without requiring tools. Other quick-release devices, for example any known latches, may alternatively be used for tool-less attachment of separation of a tabletop to/from a table base. First and second plates (550) and (600), respectively, may have any size and shape. [0168] The components of the tabletop assembly may be provided as a kit for assembly into the tabletop assembly. As shown in FIG. 12 and FIG. 13, box (2000) has a filler (fixed Styrofoam shapes) (2100) that house each base (20 and 30) on its side. The post (10) is located between the two bases, but separated by the fixed Styrofoam. The fold on the right side and the crinkles on the left side of the post are the wrapping to protect the post during shipping. FIG. 13 shows the package with the post removed and placed outside the box. Underneath the post is a hardware bag which contains the base feet, the nuts for attaching the two bases and the tabletop mount platform (410) to the respective end of the post and screws (FIG. 15) for attaching the tabletop mount platform (410) to the tabletop. Also in this package is the cover (460). The table top mount platform and cover are in a bag inside the package box.

[0169] While various embodiments incorporating the present invention have been described in detail, further modifications and adaptations of the invention may occur to those skilled in the art. However, it is to be expressly understood that such modifications and adaptations are within the spirit and scope of the present invention, which is not limited by the disclosure of the preferred embodiment. Instead, the invention should be determined entirely by reference to the claims that follow.

Claims

1. A kit for a table base assembly, comprising: a bottom base (20) having a bottom base bottom surface (270), a bottom base top surface (320), a bottom base longitudinal axis (100), a bottom base center member (110) having a plurality of bottom base center member through-holes (120), a first bottom base linear member (80) and a second bottom base linear member (90) extending in opposite directions from the bottom base center member along the bottom base longitudinal axis, and a bottom base shoulder (150) extending downwardly from the bottom base bottom surface; a top base (30) having a top base bottom surface (310), a top base top surface (315), a top base longitudinal axis (200), a top base center member (210) having a plurality of top base center member through-holes (220), a first top base linear member (180), a second top base linear member (190) extending in opposing directions from the top base center member along the top base longitudinal axis, and a first top base shoulder (290) and a second top base shoulder (300) extending downwardly from the top base bottom surface except for a first top base gap (250 and a second top base gap (260) on opposed sides of the top base center member; a post (10) with a post center longitudinal axis (11) having a post top end (70) and a post bottom end (50) opposite the post top end, wherein an angle formed by the top base longitudinal axis and the bottom base longitudinal axis is in a range of 1 Degree to 89 Degrees when at least a portion of the plurality of bottom base center member through-holes are to align with at least a portion of the plurality of the top base center member through-holes when the top base bottom surface is placed onto the bottom base top surface; and wherein the post bottom end is configured to be mounted on the top base top surface at the top base center member so that the post center longitudinal axis intersects with the bottom base longitudinal axis and the top base longitudinal axis thereof.

2. The kit of claim 1, further comprising at least two bottom thread members (40) configured to

- pass through the top base center member through-holes and the bottom base center member through-holes when the top base bottom surface is placed onto the bottom base top surface.
3. The kit of claim 2, wherein the at least two bottom thread members (40) are threaded studs secured to the post bottom end, extending from the post bottom end and are configured to pass through the top base center member through-holes.
 4. The kit of claim 2, wherein the at least two bottom thread members (40) are selected from the group consisting of threaded rods and bolts.
 5. The kit of claim 2, wherein the post bottom end includes at least two post bottom end threaded holes (570) and the at least two bottom thread members are configured to be mated to a corresponding post bottom end threaded hole after passing through the top base center member through-holes and the bottom base center member through-holes when the top base bottom surface is placed onto the bottom base top surface.
 6. The kit of claim 1, further comprising a tabletop mount platform (410) having a tabletop mount platform planar top surface (540), a tabletop mount platform bottom surface (420) with a tabletop mount platform center member (430) having at least two tabletop mount platform center member through-holes (530), and a plurality of tabletop mount platform holes (440) through the tabletop mount platform planar top surface, wherein the tabletop mount platform center member is configured to mate to the post top end.
 7. The kit of claim 2, further comprising a tabletop mount platform (410) having a tabletop mount platform planar top surface (540), a tabletop mount platform bottom surface (420) with a tabletop mount platform center member having at least two tabletop mount platform center member through-holes (530), and a plurality of tabletop mount holes through the tabletop mount platform planar top surface, wherein the tabletop mount platform center member is configured to mate to two post top end thread members (60) configured to pass through the at least two tabletop mount platform center member through holes.
 8. The kit of claim 7, wherein at least two post top end thread members are threaded studs secured to the post top end, extending from the post top end.
 9. The kit of claim 7, wherein at least two post top end thread members are selected from the group consisting of threaded rods and bolts.
 10. The kit of claim 7, wherein at least two post top end thread members are configured to be mated to a post bottom end threaded hole after passing through the tabletop mount platform center member through-holes.
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